

Artificial Neural Network Simulation in LabVIEW and MATLAB

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Artificial neural network, in essence, is an attempt to simulate the brain. When the user enters the inputs (say, p_1 , p_2 and p_3) and the expected corresponding outputs (say, t_1 , t_2 and t_3) in the program, the program trains the system and gives a final weight. The final weight is computed to get the final expected output. This program is an attempt to understand the basics of artificial neural network and how one can use it for further applications.

Artificial neural network

Artificial neural network theory revolves around the idea that certain key properties of biological neurons can be extracted and applied to simulations, thus creating a simulated (and very much simplified) brain. The first important thing to understand is that the components of an artificial neural network are an attempt to recreate the computing potential of the brain. The second important thing to understand, however, is that no one has ever claimed to simulate anything as complex as an actual brain. Whereas the human brain is estimated to have of the order of ten to a hundred billion neurons, a typical artificial neural network is not likely to have more than a thousand artificial neurons.

In theory, an artificial neuron (often called a 'node') captures all the important elements of a biological one. Nodes are connected to each other and the strength of these connections is normally given by a numeric value between -1.0 (for maximum inhibition) and +1.0 (for maximum excitation). All values between the two are acceptable, with higher-magnitude values indicating a stronger connection strength. The

transfer function in artificial neurons, whether in a computer simulation or actual microchips wired together, is typically built right into the node's design.

A transfer function (also known as the network function) is a mathematical representation, in terms of spatial or temporal frequency, of the relationship between the input and output of a linear time-invariant system.

Perhaps the most significant dif-

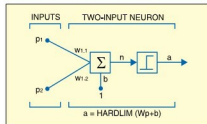


Fig. 1: Input block diagram

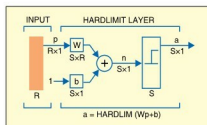


Fig. 2: Hardlimit representation

ference between artificial and biological neural nets is their organisation. While many types of artificial neural nets exist, most are organised according to the same basic structure. There are three components to this organisation: a set of input nodes, one or more layers of 'hidden' nodes, and a set of output nodes. The input nodes accept information, and are akin to sensory organs. Whether the information is in the form of a digitised picture, a series of stock values or just about any other form that can be numerically expressed, this is where the net

gets its initial data. The information is supplied as activation values, that is, each node is given a number, with higher numbers representing greater activation.

This is just like human neurons, except that rather than conveying their activation level by firing more frequently—as biological neurons do—artificial neurons indicate activation by passing this activation value to connected nodes. After receiving this initial activation, the information is passed through the network. Connection strengths, inhibition/excitation conditions and transfer functions determine how much of the activation value is passed on to the next node.

Each node sums the activation values it receives, arrives at its own activation value, and then passes that along to the next nodes in the network (after modifying its activation level according to its transfer function). Thus the activation flows through the net in one direction—from input nodes, through the hidden layers, until eventually the output nodes are activated.

If a network is properly trained, this output should reflect the input in some meaningful way. For instance, when a gender recognition net is presented with a picture of a man or woman at its input node, it must set an output node to 0.0 if the picture depicts a man or to 1.0 for a woman. In this way, the network communicates its knowledge to the outside world.

How they learn

Having explained that connection strengths are storehouses of knowledge in neural net architectures, it should come as no surprise that learn-

